The cut paradox in two signatures: sections of a solid and instants of a motion

Test 1 of I003, drafted 2026-09-08 by the independent Claude Fable 5.1 session. Earlier scripts are historical artifacts; current verification uses written proof and source review. The proposition below is a working derivation awaiting review and a prior-art audit; no claim ID is assigned. The 2026-09-09 review preserves the scalar rigidity candidate; the matrix extension and readout-noise test below retain the corrected I003 boundaries. The cone/arrow connection below is a modern heuristic; analytic continuation relates the specified evolution equations, rather than establishing a historical identity.

Result. The Euclidean statement C032, that a position-only law with ballistic support and independent stationary increments is a deterministic drift, has a Lorentzian counterpart: a scalar, translation-invariant, strongly continuous unitary evolution on the line whose kernels are supported in the light cone $[-ut, ut]$ is a uniform translation at some speed $|a| \le u$, up to a constant phase. In both signatures a bounded-speed law with nontrivial fluctuation or dispersion needs a second state variable, the velocity sign of the telegraph process or the second component of the Dirac evolution, and the continuation $\lambda \to i\omega_0$ of A09 carries the one situation into the other. The obstruction sits at the Compton wave number: the scalar reduction of the massive evolution has dispersion $\omega(k) = \sqrt{u^2k^2 + \omega_0^2}$ with branch points at $k = \pm i\omega_0/u$, which at $u = c$, $\omega_0 = mc^2/\hbar$ is $k = \pm imc/\hbar$. This is the reading of the user's remark 2 in I003 that the present calculation supports; remark 3, an h that controls convergence, is the crossover $g(z)$ of the coordinator's exact bridge note.

1. The two paradoxes as slicing laws

Cone. A solid is a family of sections $A(z)$, $0 \le z \le h$. The cone $A(z) = \pi r^2(1 - z/h)^2$ has $A'(z) = -2\pi r^2(1 - z/h)/h \ne 0$ below the apex; the cylinder has $A' = 0$. Democritus asks whether adjacent sections are equal or unequal, Chrysippus answers neither, and the continuum answer is that $A(z + \delta) - A(z) = A'(z)\delta + O(\delta^2)$: adjacent sections differ by a quantity that vanishes with the cut spacing while their ratio of differences to spacing does not. The Mohist Canon stops the halving at an endpoint; Liu Hui and Archimedes cut "until it cannot be cut" and keep the limit. Cutting the solid at more places preserves the old sections; this is the restriction consistency of a fixed solid, the analogue of C034 for a fixed bridge.

Arrow. A motion is a family $X(t)$. Zeno asks whether at each instant the arrow is where it is, and Aristotle answers that the argument takes time to be composed of nows (Physics VI.9). In the project's models the state at a cut is (X, V): the finite-speed bridge of C033 retains the velocity sign at every cut, and its midpoint atom is exactly the arrow at its turning instant, where the position is $uT/2$ with probability $1/I_0(\lambda T)$ and the velocity is fixed only by the right-continuous convention. Al-Naẓẓām's leap is the discrete crossing of the same instant.

Under the exchange of the slicing variable z with t and of the slope A' with the velocity, the two dilemmas are one question: what an infinitesimally adjacent slice carries. The continuation of A09 makes the exchange precise for the telegraph and checkerboard dynamics, and the rest of this note states what survives on each side.

2. The Euclidean face

C032. If $\mu_0 = \delta_0$, $\mu_{s+t} = \mu_s * \mu_t$ and $\operatorname{supp}\mu_t \subset [-ut, ut]$, then $\operatorname{Var}\mu_T = n \operatorname{Var}\mu_{T/n} \le u^2T^2/n \to 0$ and $\mu_t = \delta_{bt}$ with $|b| \le u$. Adjacent sections of such a law are translates of one another. The telegraph process escapes the theorem because its position alone is not Markov: its kernel's Fourier transform $e^{-\lambda t}[\cosh(st) + (\lambda/s)\sinh(st)]$ with $s = \sqrt{\lambda^2 - u^2k^2}$ is an even function of s, hence entire in k, but it is not a convolution semigroup in t.

3. The Lorentzian face

Proposition. Let $(U_t)_{t\in\mathbb{R}}$ be a strongly continuous one-parameter unitary group on $L^2(\mathbb{R})$ that commutes with translations, and suppose that for every t the kernel K_t with $U_t\psi = K_t * \psi$ has support in $[-u|t|, u|t|]$. Then there are real constants a, b with $|a| \le u$ such that $U_t\psi(x) = e^{-ibt}\psi(x - at)$.

Proof. Commutation with translations makes U_t a Fourier multiplier by a unimodular measurable $m_t(k)$, and Stone's theorem gives $m_t(k) = e^{-it\omega(k)}$ for a real measurable ω. The support hypothesis and the Paley–Wiener–Schwartz theorem extend each m_t to an entire function with $|m_t(k)| \le C_t(1 + |k|)^N e^{u|t||\operatorname{Im} k|}$. The identity $m_t m_{-t} = 1$ holds on the real axis, hence on $\mathbb{C}$, so m_t has no zeros and $G_t = \log m_t$ is entire. For fixed t, the bounds for m_t and m_{-t} give $|\operatorname{Re} G_t(z)| \le u|t||\operatorname{Im} z| + N_t \log(1 + |z|) + C_t$. Harmonic derivative estimates make this function affine; its value on the real axis is zero. Thus $m_t(k) = e^{-i(a_t k + b_t)}$, with real a_t, b_t and $|a_t| \le u|t|$. The group law makes a_t additive; the bound implies $a_t = at$. Strong continuity makes the remaining phase a continuous character, hence e^{-ibt}. This proves the form without selecting logarithms continuously in time. □

Reading. Scalar evolutions in this class are translations at any $|a| \le u$, with an arbitrary constant phase rate. The massless chiral choices $a = \pm u$ saturate the speed bound. Interpreting phase rate as energy requires an action conversion factor. The Dirac evolution keeps light-cone support and unitarity by carrying two components; its scalar reduction has $\omega(k) = \sqrt{u^2k^2 + \omega_0^2}$, which is real on the axis but has branch points at $k = \pm i\omega_0/u$ and so cannot be the dispersion of a scalar kernel of compact support. At $u = c$ and $\omega_0 = mc^2/\hbar$ the branch points sit at the inverse reduced Compton length. The analytic obstruction and the physical crossover Δ_* of the P02 note are the same scale seen from the two sides of the continuation.

4. The double limit

Refining cuts in time and resolving positions together is the double limit of I003's remark 2. At finite speed and fixed λ, the window observable obeys $\mathsf{h}(\Delta) \le H_*\Delta/\Delta_*$ (P02), and the conditioned midpoint obeys the exact law $\kappa_{\mathrm{mid}}/H_* = g(\lambda T) = zR(z)[1 - R(z)]$ of the bridge-crossover note, with cubic onset. The fixed-preparation short-window intrinsic coefficient vanishes under a speed ceiling; the specified crossover window $\Delta_* = K/(mu^2)$ concerns a long-window plateau, and g describes its approach. This does not assert a limit for noisy records whose error scale changes with the cut. Which premise makes the plateau positive and common to all bodies is A08's question, not a consequence of scalar finite propagation.

5. The matrix boundary and a specified noisy double limit

The scalar proposition is **not** a classification of matrix-valued finite-speed semigroups. A candidate, still unproved, says that a translation-invariant, strongly continuous contraction semigroup on $L^2(\mathbb{R}; \mathbb{C}^n)$ whose convolution kernels have support in $[-ut, ut]$ has Fourier multipliers $e^{t(B - ikA)}$, where $A = A^\dagger$, $\|A\| \le u$ and $B + B^\dagger \le 0$. The missing steps include pointwise time continuity from strong operator continuity, a justified matrix logarithm and differentiation of its generator. Positivity alone does not make B a conservative Markov switching generator: $T_t f = e^{-t} f$ is positive and has zero propagation but loses mass. Conservation must be required separately. The scalar proof above does not discharge these obligations.

For the symmetric two-state telegraph generator $Q = \lambda(\sigma_x - I)$ and $V = u\sigma_z$, the exact matrix identity is $\|[V, Q]\| = 2u\lambda$, while the stationary velocity plateau is $H_* = mu^2/\lambda = 2mu^3/\|[V, Q]\|$. A coherent two-component model with $A = c\sigma_z$, $B_H = \omega_0\sigma_x$ similarly has $\|[A, B_H]\| = 2c\omega_0$. Converting frequency into rest energy uses a **supplied** action K, $m = K\omega_0/c^2$; physical $\hbar$ enters only if separately identified with K. A commutator norm does not see every slow mode: the four-state counterexample keeps the plateau fixed as an independent

hidden label's relaxation gap closes. For $Q = 0$, a random constant velocity $\pm u$ instead has $\mathrm{Var}(X_\Delta - X_0) = u^2\Delta^2$; a deterministic velocity has zero variance. Commuting velocity sectors therefore do not justify the old general zero-response claim.

There is also an **exact, model-dependent readout effect**. Define $\mathsf{h}(\Delta) = m\,\mathrm{Var}(X_\Delta - X_0)/\Delta$ for a fixed bounded-speed motion. At the two endpoints add independent centered errors, independent of the motion, each of variance $\varepsilon^2/12$. The observed increment then satisfies

$$\mathsf{h}_\varepsilon(\Delta) = \mathsf{h}(\Delta) + \frac{m\varepsilon^2}{6\Delta}. \tag{I003}$$

For fixed $\varepsilon > 0$, the $\Delta \downarrow 0$ limit is infinite; for fixed $\Delta > 0$, removing the readout error first and then refining gives zero by finite speed. Along $\varepsilon^2 = 6\kappa_0\Delta/m$, the observed coefficient tends to any chosen $\kappa_0 \geq 0$. Independent dither or additive-noise measurement is a premise here; fixed-lattice deterministic rounding need not satisfy it. On a uniform N-cell partition, pin endpoint errors to zero and give the $N - 1$ interior errors this variance. Expanding the kinetic-action squares, the expected added action is $m(N-1)\varepsilon^2/(12\Delta)$, with correlated neighboring increment errors accounted for in the expectation. This scaling selects a **readout bias**, not a lower bound on intrinsic action and not physical $\hbar$.

6. Cone source trail and the free-kernel comparison

The historical H04 witness locates the cone dilemma in Xylander's 1570 *Moralia*, pp. 823–824. Its reception search did not verify a specific early-modern mathematician using that passage in work on indivisibles; Auffret's separate rolling-cone/Leibniz example does not prove that transmission. Rivero's 1999 cone essay motivates keeping the finite separation and rescaled difference apart. Rivero's 1998 path-integral proposal motivates eliminating intermediate coordinates; Brouder's rooted-tree algebra and Cariñena et al.'s tangent-groupoid proposal are separate nonlinear and quantum-completion leads. The retained time-refinement manuscript and PDF give their source reading and the full normalized free-kernel proof.

In that proof a positive Gaussian kernel of variance $\kappa t/m$ composes exactly under integration over an inserted position for every $\kappa > 0$. Weak continuity and Gaussian time homogeneity permit every $\kappa = ma \geq 0$; $a = 0$ is the deterministic Dirac member. Requiring a positive density rules out zero by premise. Passing from positive kernels to oscillatory unitary propagation is another chosen step, and neither step identifies κ with physical $\hbar$. The Newton synthesis distinguishes this free fixed-line result from the two-regulator RG question.

7. Consequence for STATE

Finite-speed scalar rigidity, a corrected matrix conjecture and the specified readout double limit are distinct statements. Intrinsic fluctuations vanish under fine bounded-speed cuts; independent errors can create an arbitrary resolution-dependent coefficient. Neither phenomenon supplies a universal positive physical action. The scalar proof remains a working unrefereed derivation pending its stated Paley–Wiener and harmonic-growth audit; the matrix claim remains open. The ancient parallel is a modern heuristic, not a transmission claim or a quantum reconstruction.